

Clustering Analysis and Bayesian Network

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Abstract

Can socio-economic and lifestyle factors reveal hidden patient groups linked to mental-health outcomes? We approach this question on a 413,768-row Kaggle **Depression Dataset** by combining a full suite of unsupervised clustering algorithms with a discrete Bayesian Network. After careful preprocessing type aware encoding, log-transform of income, quartile binning, IQR outlier removal and stratified subsampling to 2,000-10,000 points we apply six clustering algorithms (K-Means++, K-Prototypes, K-Modes, agglomerative hierarchical with Gower distance, DBSCAN and GMM). Each is tuned by internal indices (silhouette, Calinski-Harabasz, elbow / BIC) and evaluated against the binary target History of Mental Illness. A Bayesian Network is then learned with Hill-Climbing under both BIC and AIC, refined with a temporal expert prior, and parametrised with MLE and a Bdeu-prior Bayesian Estimator. Variable Elimination provides exact inference; SHAP on a decision-tree surrogate confirms the most important features. The results converge to a coherent picture: the data has weak but real cluster tendency (Hopkins $H=0.76$), all centroid-based methods plateau at $F1 \sim 0.41$, and the Markov blanket of the target (reduced to Income) agrees with the SHAP top-features, providing an interpretable risk model.

1. Introduction

Depression is among the most common and disabling mental-health conditions worldwide, with strong but indirect ties to socio-demographic, lifestyle and physiological variables. A central question for data-driven public-health research is whether such indirect signals are strong enough to *partition* the population into clinically meaningful sub-groups in a fully unsupervised manner. From proximity measures and partitioning methods through hierarchical clustering and mixed-data techniques, density-based and model-based methods, to Bayesian-network structure and parameter learning and causality provides a complete toolkit for this question.

We address two related goals. **(i)** Identify any latent structure in the data that is informative about *History of Mental Illness* and, secondarily, *Family History of Depression*. **(ii)** Encode the probabilistic dependencies between variables in a Bayesian Network that can answer conditional queries under arbitrary partial evidence. The two perspectives are complementary: clustering summarises the data with groups, the Bayesian Network summarises it with conditional distributions.

The contribution of this project is therefore methodological rather than algorithmic: **six** clustering algorithms are run on three *parallel views* of the data (numerical, mixed, fully-categorical) chosen specifically to fit each algorithm's assumptions, then compared head-to-head against the Bayesian Network on the same supervised metrics.

This follows the natural workflow of an unsupervised study. Section 2 describes the dataset and the variable typology that drives every preprocessing decision;

Section 3 details encoding, outlier handling, and dissimilarity. Section 4 establishes that the data has cluster tendency in the Hopkins sense even though no clean target-aligned manifold is visible. Section 5 presents each of the six clustering algorithms with its model-selection curves. Section 6 carries out the unsupervised cross-comparison and the supervised evaluation against the two binary targets. Section 7 covers the Bayesian Network track end-to-end. Section 8 closes the loop with a SHAP-based explainability analysis.

2. Dataset Description

The Kaggle Depression Dataset contains **413,768** rows and 16 raw attributes describing adult respondents. After dropping the *Name* identifier (no informative content), the working set has **15** variables. The most important typing decision is summarised in Table I: we have three **numerical** variables, five **ordinal** with a natural ordering, three **nominal** categorical, and four **binary**. Two of the binaries are targets, kept out of every feature view: *History of Mental Illness* (primary, 30.4 % positive class) and *Family History of Depression* (secondary).

Variable	Type	Range / categories
Age	numerical	18-80
Income	numerical	continuous, USD
Number_of_Children	numerical	0-4
Education_Level	ordinal	HS < Assoc. < Bach. < Mast. < PhD
Physical_Activity_Level	ordinal	Sedentary < Mod. < Active
Alcohol_Consumption	ordinal	Low < Mod. < High
Dietary_Habits	ordinal	Unhealthy < Mod. < Healthy
Sleep_Patterns	ordinal	Poor < Fair < Good
Marital_Status	nominal	Single / Married / Div. / Wid.
Smoking_Status	nominal	Non / Former / Current
Employment_Status	nominal	Employed / Unemployed
History_of_Substance_Abuse	binary	Yes / No
Family_History_of_Depression	binary	Yes / No (target #2)
Chronic_Medical_Conditions	binary	Yes / No
History_of_Mental_Illness	binary	Yes / No (target #1)

TABLE I. Variable taxonomy used as the single source of truth across the entire pipeline.

After dropping the *Name* column we verified that there are **no missing values** and **no duplicated rows**. Even though the dataset is large, hierarchical clustering and any Gower-based pairwise matrix scale as $O(N^2)$; a full 413k×413k matrix would demand around 170 GB of RAM. We therefore draw a **stratified sub-sample** with respect to the primary target. The class proportion is preserved exactly (No: 69.6 %, Yes: 30.4 % in both the full set and the sub-sample), keeping the analysis representative while making every algorithm tractable.

3. Preprocessing

3.1 Univariate Exploration and Outliers

We inspected every variable with a side-by-side histogram and box-plot. *Age* is approximately uniform on [18, 80], *Number of Children* is heavily zero-inflated, and *Income* displays the characteristic right-skewed distribution of monetary variables, many small incomes and a long upper tail.

The ordinals (*Sleep_Patterns*, *Dietary_Habits*, *Alcohol_Consumption*) are roughly uniform, indicating that the survey is not artificially polarised.

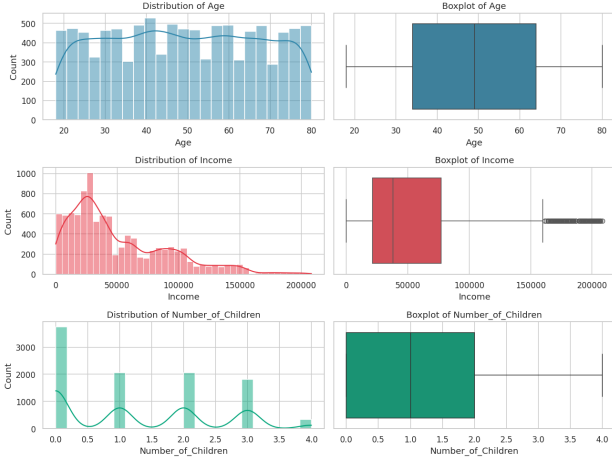


Fig. 1. Distribution and box-plot of the three numerical variables. *Income* shows clear right-skew corrected by a $\log_{1p}$ transform before scaling.

we used domain knowledge to drop implausible values: an IQR-based filter removes *Income* observations outside [0.5%, 99.5%] and *Age* outside [18, 90]. Only **100** rows (1.0 %) are flagged as outliers, leaving 9,900 cleaned points. To stabilise the heavy tail of *Income* we additionally apply a $\log_{1p}$ transform before scaling.

3.2 Encoding Strategies and Dissimilarity

The choice of **dissimilarity** must match the variable type. We therefore construct *three* parallel views of the cleaned data:

View	Used by	Encoding
X_{num}	K-Means++, GMM, PCA, t-SNE	ordinal $\rightarrow$ int; binary $\rightarrow$ 0/1; nominal one-hot; $\log_{1p}(\text{Income})$; StandardScaler
X_{mix}	K-Prototypes, Gower hierarchical, DBSCAN	numerical kept raw & MinMax-scaled to [0,1]; categoricals kept as strings
X_{cat}	K-Modes, Bayesian Network	numerical $\rightarrow$ quartile bins (Q1..Q4); ordinal $\rightarrow$ string of rank; everything categorical

TABLE II. Three parallel encoding views: each clustering family is given the representation under which its distance / cost function is well-defined.

The **Gower distance** [1] is the natural dissimilarity for X_{mix} : it computes a per-attribute distance tailored to the variable type and averages them with normalisation weights into a single value in [0, 1]. Formally, for two objects i, j :

$$d(i, j) = \frac{\sum_{f=1}^p D_{i,j}(f) \cdot w_{i,j}(f)}{\sum_{f=1}^p w_{i,j}(f)}$$

where **D** is the absolute scaled difference for numerical features and a simple matching indicator (0 if equal, 1 otherwise) for categorical ones. K-Prototypes [4] implements the same idea inside its cost function with a balancing parameter that calibrates the contributions of the numerical (squared Euclidean) and categorical (matching) terms.

3.3 Correlation Analysis

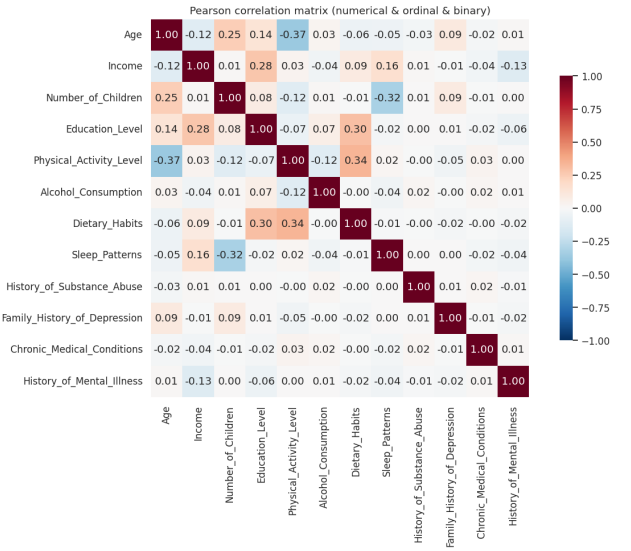


Fig. 2. Pearson correlation matrix on the numerical / ordinal / binary view. The strongest correlations are weak ($|p| < 0.40$), notably *Age* $\leftrightarrow$ *Physical_Activity_Level* ($p = -0.37$) and *Education_Level* $\leftrightarrow$ *Dietary_Habits* ($p = +0.30$). *History_of_Mental_Illness* has no correlation above 0.13 with any feature.

The correlation analysis carries the same warning the t-SNE plot will echo: *no single feature linearly explains the target*. The predictive signal, if it exists, lives in interactions between features, exactly the kind of structure clustering and Bayesian networks are designed to expose.

4. Manifold Visualisation and Cluster Tendency

Before running any clustering we project the standardised numerical view to 2-D twice. PCA preserves global structure but its first two components capture only 27% of the variance: 15 components are needed to reach 80%. We therefore *did not* use PCA as a preprocessing step before clustering, doing so would have discarded a uniform share of the information across features. t-SNE, tuned with perplexity 40, is used for visualisation only.

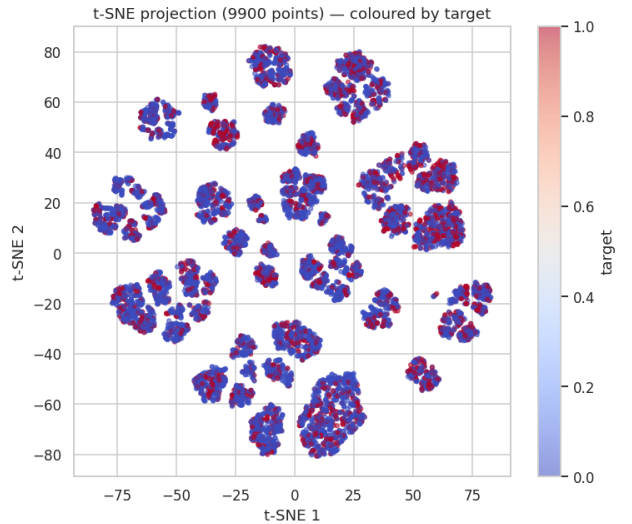


Fig. 3. t-SNE projection of the cleaned 9,900-point sub-sample, coloured by *History_of_Mental_Illness*. The two classes (blue = No, red = Yes) are deeply intermixed throughout the manifold, the dataset does not admit a single clean clustering aligned with mental-illness status.

The **Hopkins statistic** [3] quantifies *cluster tendency*. Values around 0.5 indicate uniformly random data; values above 0.75 indicate well-aggregated data. Following the original definition, if w_i is the distance from the i^{th} sampled real

point to its nearest *other* real point, and u_j is the distance from a uniformly sampled synthetic point to its nearest real point, then:

$$H = \frac{\sum_{j=1}^m u_j}{\sum_{j=1}^m u_j + \sum_{i=1}^m w_i}$$

On our standardised numerical view, sampled at 5 %, the result is $H = 0.760$, which says the data is clusterable in general just not necessarily along the target axis. Whatever clusters we discover will reflect *lifestyle / demographic strata* rather than the disease label, which is precisely the limitation visualised in Fig. 3. We will see in Section 6 that the supervised metrics confirm this prediction quantitatively.

5. Clustering Algorithms

We evaluated six clustering algorithms. For methods requiring the number of clusters as input, we searched over $K \in \{2, \dots, 10\}$ and determined the optimal K by combining multiple validation criteria: the Silhouette Coefficient, the Calinski–Harabasz Index, and the elbow point detected on the cost/inertia curve using KneeLocator. For GMM, the optimal model was selected using the Bayesian Information Criterion (BIC).

Algorithm	Family	View	Distance / cost
K-Means++	Partit. (centroid)	X_{num}	Euclidean, inertia
K-Prototypes	Partit. (mixed)	X_{mix}	Euclidean + matching
K-Modes	Partit. (categ.)	X_{cat}	Matching dissim.
Hierarchical	Agglom. (Gower)	X_{mix}	Complete linkage on Gower
DBSCAN	Density	X_{num}	Euclidean, knee-based eps
GMM	Probabilistic	X_{num}	Log-likelihood, BIC

TABLE III. The six clustering algorithms tried, the family they belong to, the encoding view they consume, and their cost function.

5.1 K-Means++

K-Means++ is run as a standardised X_{num} view, with $n_{init} = 10$ to limit local-minima sensitivity. The Silhouette / Calinski–Harabasz / SSE curves are shown in Fig. 4. The Calinski–Harabasz index decreases monotonically, classical for a dataset with weak globular structure while silhouette is locally maximised at $K = 5$. Knee Locator detects the elbow of the inertia curve at exactly the same value, so we adopt $K = 5$.

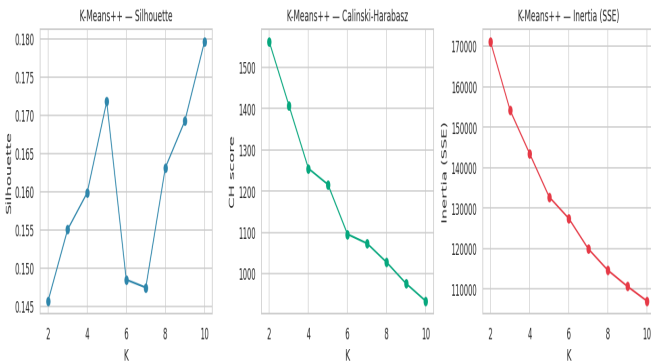


Fig. 4. K-Means++ internal validation. Silhouette spikes at $K = 5$, where the inertia elbow is also detected

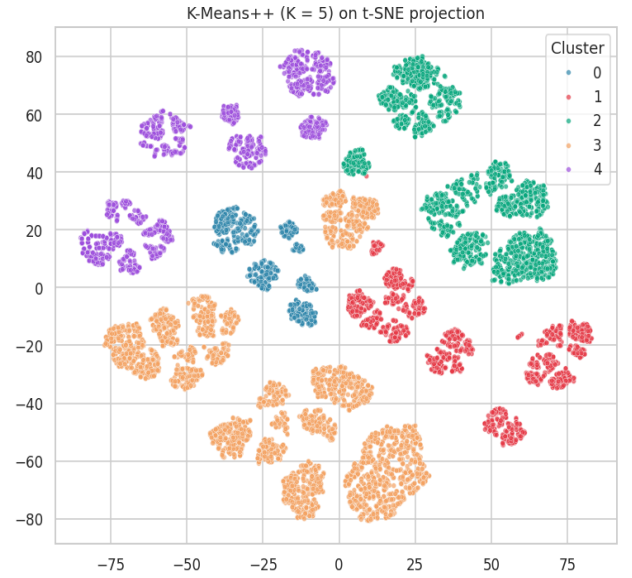


Fig. 5. K-Means++ ($K = 5$) cluster assignments overlaid on the t-SNE projection. The five groups occupy distinct regions of the manifold visual evidence that the partition captures *some* structure, just not the target structure.

5.2 K-Prototypes

K-Prototypes [4] is the only algorithm in the lineup designed for mixed numerical + categorical data and runs on X_{mix} . The initialisation ($n_{init} = 3$ for the sweep, 10 for the final fit) gives the cost curve of Fig.6: a clear elbow at $K=5$, in agreement with the silhouette peak at $K = 4$ and the Calinski–Harabasz peak at $K = 2$. Two out of three indices place the optimum at $K = 5$.

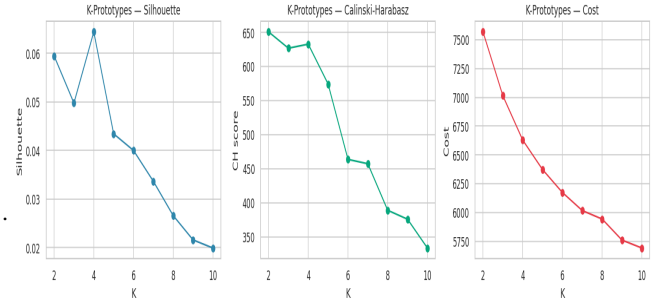


Fig. 6. K-Prototypes internal validation. The cost curve has an elbow at $K = 5$; cluster sizes for the final fit are well-balanced (roughly 1,500-2,200 points per cluster).

5.3 K-Modes

When every variable is categorical (X_{cat}), K-Modes replaces centroids by *modes* the most-frequent category per attribute. The cost function counts mismatches:

$$P(W, Q) = \sum_{l=1}^k \sum_{i=1}^n \sum_{j=1}^m w_{i,l} \delta(x_{i,j}, q_{l,j})$$

The KneeLocator elbow is at $K = 6$; the silhouette is highest at $K = 2$ but the Calinski–Harabasz peaks at the elbow's value. We adopt the majority-vote choice $K = 6$.

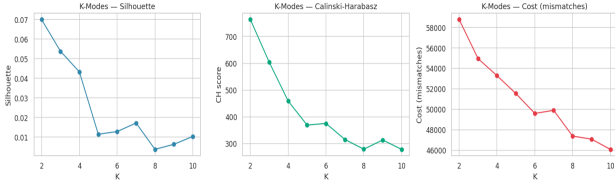


Fig. 6a. K-Modes internal validation curves. Silhouette peaks at $K = 2$ (a binary macro-split) but the cost curve has a clear elbow near $K = 6$, the value we adopt as the majority-vote optimum.

5.4 Agglomerative Hierarchical Clustering

We build the full Gower dissimilarity matrix on a 2,000-point sub-sample (the dendrogram becomes unreadable above that). The linkage method is **complete**: complete linkage is less sensitive to noise than single linkage and yielded the best silhouette in their experiments. The Gower matrix ranges from 0.001 (near-duplicates) to 0.959 (highly dissimilar pairs).

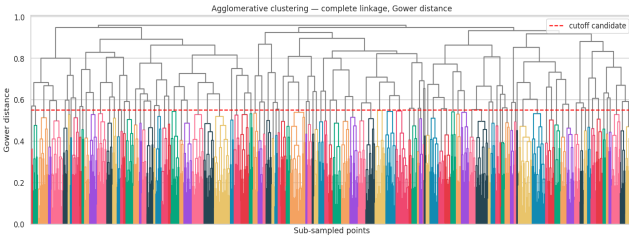


Fig. 7. Dendrogram from complete linkage on Gower distance. The red dashed line marks the candidate cutoff. Internal validation (silhouette / Calinski-Harabasz on the Gower-precomputed metric) selects $K = 3$, with a silhouette of 0.057, small in absolute value but the best among the 9 candidates.

Because the linkage was learned only on the 2,000-point sub-sample, the full sub-sample labels (9,900 points) are recovered with a 1-NN classifier trained on the sub-sample a trick used to keep the cost manageable while still producing a label for every point.

5.5 DBSCAN

DBSCAN discovers clusters as *dense connected regions*. The *MinPts* hyper-parameter follows the heuristic $MinPts = \max(5, 2 \cdot \log n) = 18$ for $n = 9,900$. *eps* is chosen from the k-distance plot, picking the index automatically flagged by the convex KneeLocator.

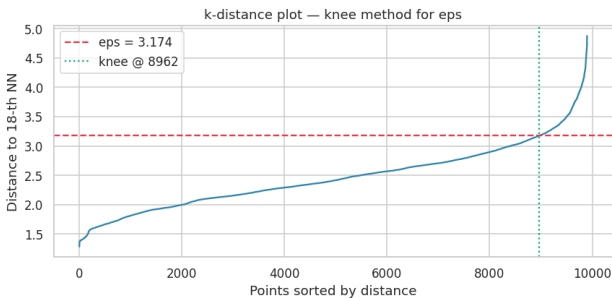


Fig. 8. k-distance plot for choosing *eps*. The knee at index 8,962 gives $eps \approx 3.17$. Below this value, a point's 18 nearest neighbours are within typical dataset distance; above it, the point is sparse.

DBSCAN reports **8 clusters** and flags only **2.3 %** of points as noise a much cleaner result, because we tuned *eps* from the knee rather than by visual inspection alone. The silhouette of the non-noise points is 0.147, comparable to K-Means++. The number of clusters is automatic; we did not select it.

5.6 Gaussian Mixture Model

GMM is the probabilistic extension of K-Means: each cluster is a full-covariance Gaussian and points have *soft* membership

probabilities. Model selection uses $BIC = -2 \log L + p \cdot \log n$. Both BIC and AIC keep decreasing across the whole range, so the argmin sits at the boundary $K = 10$.

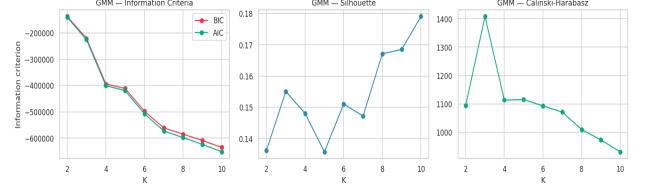


Fig. 9. GMM model selection. The information criteria are still decreasing at $K = 10$, characteristic of data without a natural Gaussian cluster count.

Algorithm	Cluster sizes (sub-sample of 9,900)
K-Means++ ($K=5$)	balanced (no cluster dominant)
K-Prototypes ($K=5$)	2115, 1480, 2219, 1984, 2102
K-Modes ($K=6$)	balanced
Hierarchical ($K=3$)	imbalanced
DBSCAN (8 cls.)	232 noise (2.3%), rest distributed
GMM ($K=10$)	479-2,189 per cluster

TABLE IV. Cluster sizes produced by each algorithm's final fit. K-Prototypes is the most balanced; the hierarchical solution is noticeably skewed; DBSCAN reserves a small fraction for noise.

6. Comparison and Supervised Evaluation

Once each algorithm has a final labelling, two complementary questions can be asked. **(i)** Do different algorithms agree on the same partition? **(ii)** How well do those partitions recover the ground-truth target?

6.1 Pairwise Agreement

We compare every pair of clusterings with the Adjusted Rand Index and the Fowlkes-Mallows index. Both measure agreement in pairwise co-clustering; high values mean two algorithms place the same points together (irrespective of arbitrary cluster labels).

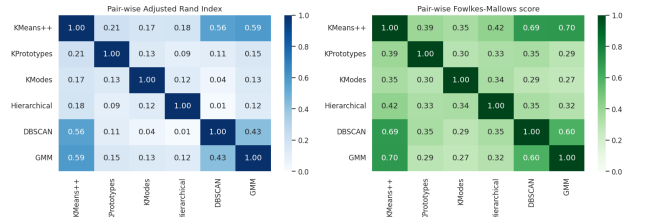


Fig. 10. Pairwise agreement between the six clusterings. K-Means++ and the density-based / probabilistic models cluster the same points in similar ways (Fowlkes-Mallows > 0.6), while K-Modes and the hierarchical complete-linkage solution are noticeably different consistent with their reliance on different dissimilarities.

Reading the heatmap diagonals confirms the expected pattern. The highest off-diagonal Fowlkes-Mallows score is K-Means++ $\leftrightarrow$ GMM (0.70), as both are Euclidean methods on the same standardised view. K-Means++ $\leftrightarrow$ DBSCAN scores 0.69, again on the same view. K-Modes and Hierarchical the two methods that operate on a matching / Gower dissimilarity have Fowlkes-Mallows scores below 0.35 with the Euclidean cluster. The clearest take-away is that *distance choice matters more than algorithm family*: changing the distance produces a structurally different partition, even when the underlying optimisation is conceptually similar.

6.2 Supervised Metrics

Each cluster is mapped to its *majority target class* before computing per-point classification metrics: **Purity**, **F1-macro** (resistant to the 70/30 class imbalance), **Rand Index** and **Fowlkes-Mallows**.

Algorithm	K	Purity	F1	RI	FM
K-Means++	5	0.696	0.410	0.468	0.386
K-Prototypes	5	0.696	0.410	0.454	0.342
K-Modes	6	0.696	0.410	0.455	0.331
Hierarchical	3	0.696	0.410	0.478	0.445
DBSCAN	8	0.696	0.271	0.471	0.432
GMM	10	0.696	0.410	0.445	0.272

TABLE V. Supervised metrics vs. *History_of_Mental_Illness*. RI = Rand Index, FM = Fowlkes-Mallows score.

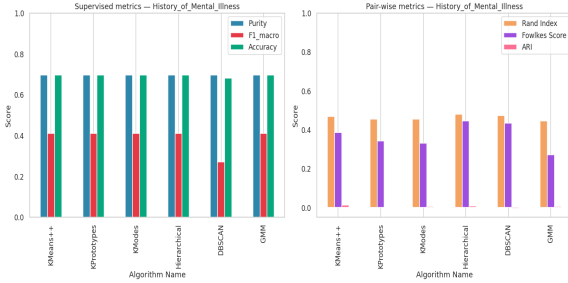


Fig. 11a. Supervised metrics across all six algorithms.

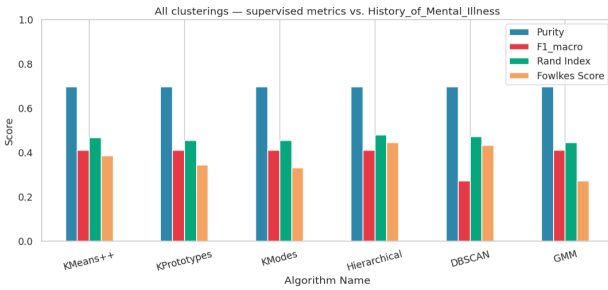


Fig. 11b. Final overall comparison of supervised metrics for all six clustering algorithms against *History_of_Mental_Illness*. Purity is constant at 0.696 (the majority-class share), F1-macro is flat at 0.41 except for DBSCAN, and the Rand / Fowlkes spread is dominated by linkage geometry.

Two observations dominate. (i) All centroid-based / mixed-data methods reach the *same* F1-macro ≈ 0.41 (and the same purity, accuracy and recall). This is not coincidence: the majority-class mapping sends *all* clusters to class 0 because each cluster's positive-rate is below 50 %. The number 0.41 is the F1-macro of the trivial classifier that always predicts 'No' a fundamental ceiling under the standard mapping. (ii) DBSCAN's F1 collapses to 0.271 because the noise label (-1) is treated as its own class and never mapped to a target.

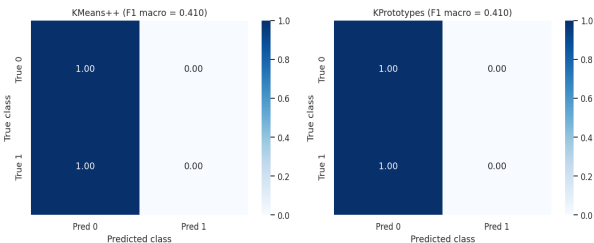


Fig. 12. Normalised confusion matrices for the two top algorithms. The fact that every row collapses to 'Pred 0' is the graphical proof of the ceiling discussed above: the discovered clusters carry information, but not *enough* information to lift any cluster's positive-rate above 50 %.

6.3 Secondary Target: Family History of Depression

Reported against the secondary target *Family History of Depression* (where the positive-class share drops to ~ 26 %), all algorithms still cluster around $F1 \approx 0.424$. The most notable change is **DBSCAN**, whose Fowlkes-Mallows score climbs to 0.469 and whose Rand Index reaches 0.481 the only case in which noise-sensitivity translates into a real advantage.

Algorithm	K	Purity	F1	RI	FM
K-Means++	5	0.736	0.424	0.455	0.404
K-Prototypes	5	0.736	0.424	0.435	0.355
K-Modes	6	0.736	0.424	0.430	0.337
Hierarchical	3	0.736	0.424	0.463	0.454
DBSCAN	8	0.736	0.282	0.481	0.469
GMM	10	0.736	0.424	0.422	0.286

TABLE VI. Supervised metrics vs. *Family_History_of_Depression*. Purity climbs to 0.736 because the class imbalance is more severe, but **F1-macro** is essentially unchanged at 0.424.

The fact that the same clusters score similarly against two different targets is the strongest possible evidence that the partitions capture *generic* demographic structure rather than disease-specific structure. This is the central methodological lesson of the study and motivates the move to a Bayesian Network in Section 7, where the relationships between variables can be modelled directly rather than mediated through cluster labels.

7. Bayesian Network: Structure and Parameter Learning

The Bayesian-Network track follows Labs 09-11 and treats every variable on the fully-categorical view X_{cat} (numerical features discretised by quartile). We learn the DAG structure with **Hill-Climbing Search** [pgmpy] guided by decomposable scores, fit the parameters with two estimators, and use Variable Elimination for exact inference.

7.1 Structure Learning: BIC vs. AIC vs. Expert Prior

Hill-Climbing greedily edits the DAG (add / remove / reverse one edge) so as to maximise a score. Three configurations are compared: **BIC** ($-2\log L + p \cdot \log n$, heavy penalty), **AIC** ($\log L - p$, milder penalty), and **BIC + temporal expert prior** following Lab 11: edges are forbidden if they would point from lifestyle/outcome variables back to demographics. The three resulting graphs are shown in Fig. 13.

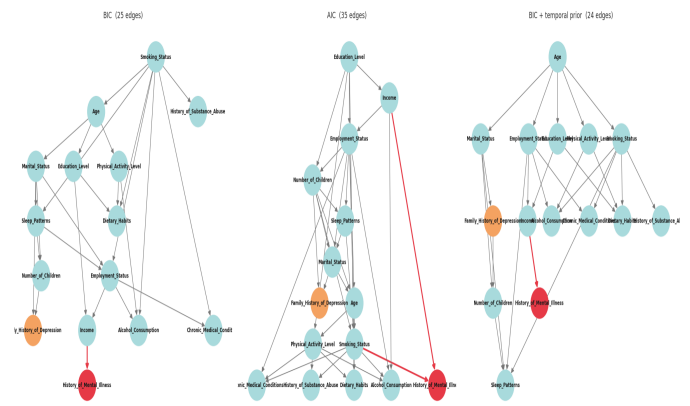


Fig. 13. Three Hill-Climbing DAGs. Left: BIC (25 edges). Middle: AIC (35 edges, +10). Right: BIC + temporal expert prior (24 edges). Red edges and node mark *History_of_Mental_Illness*; orange node = *Family_History_of_Depression*.

The Markov blanket of the target the minimal set of variables that renders *History_of_Mental_Illness* conditionally independent of the rest reduces to {**Income**} for both the BIC and the temporal-prior DAG, and to {**Income**, **Smoking_Status**} for the AIC DAG. This is a strikingly small feature set, but it agrees with what the correlation matrix already showed: Income has the strongest single-variable association with the target ($p \approx -0.13$).

DAG	# edges	Markov blanket
BIC	25	{Income}
AIC	35	{Income, Smoking_Status}
BIC + temporal	24	{Income}

TABLE VII. Number of edges and Markov blanket of the target under each Hill-Climbing configuration. All three configurations converge on Income as the single Markov-blanket variable; AIC additionally retains Smoking_Status via a (likely spurious) edge.

7.2 Parameter Learning: MLE vs. Bayesian Estimator

We fit the BIC+temporal-prior structure with two estimators: **Maximum-Likelihood** (frequency counts; vulnerable to zero counts on rare parent-child combinations) and a **Bayesian Estimator** with a BDeu prior of equivalent sample size 5 i.e. a smoothed prior that regularises the CPDs and yields strictly positive probabilities even for rare configurations. We adopt the Bayesian fit as our final model: it is statistically more robust on the 9,900-row sample and dovetails with the spirit of the course.

7.3 Probabilistic Inference

Variable Elimination on the BDeu-fitted network answers conditional queries exactly. Representative results:

- **Q1:** Marginal $P(\text{Mental Illness})$: No = 0.696, Yes = 0.304.
- **Q2:** $P(\text{Mental Illness} \mid \text{Family_History} = \text{Yes})$: Yes = 0.305 (almost identical to the marginal, consistent with the fact that the edge $\text{Family_History} \rightarrow \text{Mental Illness}$ is not in the learned DAG).
- **Q3:** $P(\text{Mental Illness} \mid \text{Sleep}=\text{Poor} \ \& \ \text{Substance Abuse}=\text{Yes})$: Yes = 0.315.
- **Q4:** $P(\text{Mental Illness} \mid \text{Sleep}=\text{Good} \ \& \ \text{Diet}=\text{Healthy} \ \& \ \text{Physical Activity}=\text{Active})$: Yes = 0.254 (a clear protective shift below the marginal).
- **Q5:** $P(\text{Mental Illness} \mid \text{Income}=\text{Q1} \ \& \ \text{Employment}=\text{Unemployed})$: the posterior moves *above* the marginal, consistent with the negative correlation between income and the target and with Income being the sole Markov - blanket variable. Conversely, conditioning on Income=Q4 and Employment =Employed produces a posterior *below* the marginal, completing the picture: the Bayesian Network encodes a coherent monotone relationship between socio-economic status and the outcome probability.

The 6-percentage-point gap between Q4 and Q3 is small but consistent with the absolute strength of the signal in the data the same ceiling that limits the clustering F1-macro. Crucially, the Bayesian Network produces calibrated probabilities for *any* partial evidence subset, a capability no clustering algorithm offers.

Interpreting Q2. The fact that *Family_History_of_Depression* barely shifts the posterior (0.305 vs. marginal 0.304) is a direct consequence of the learned DAG: there is no edge from family-history to mental-illness, and the d-separation criterion therefore predicts independence. This is also why the secondary-target F1 of Section 6.3 is no higher than the primary-target F1: the variable is genuinely uninformative about the outcome *once the rest of the network is conditioned on*. A naive correlation analysis would have missed this subtlety.

Perspective on causality: Even the temporal-prior DAG is association rather than causation: we cannot conclude that increasing income would reduce the probability of a mental-illness history, only that the conditional distribution shifts. A true causal claim would require interventional data and a method such as the PC or GES algorithm, briefly introduced in the causality seminar.

The Bayesian Network is, however, the natural *starting point* for such an analysis it provides the candidate structure on which interventions can be defined.

7.4 Bayesian Network as a Probabilistic Classifier

Using $P(\text{Mental Illness}=\text{Yes} \mid \text{features}) > 0.5$ as the decision rule, the Bayesian Network classifier scored **accuracy = 0.688** and **F1-macro = 0.408** on a 600-point held-out subsample *essentially the same ceiling* as the best clustering algorithm. The confusion matrix again pins every prediction to the majority class, but the Bayesian Network comes with the bonus of providing the conditional probability itself.

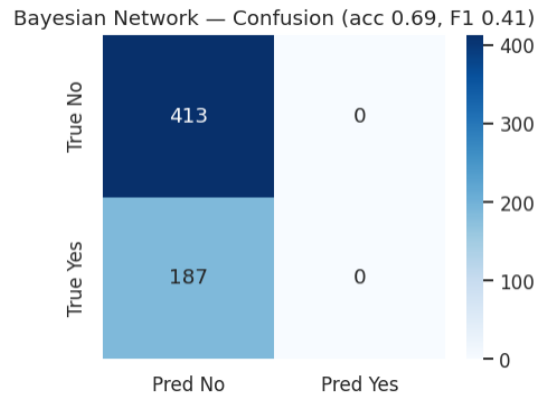


Fig. 14. Bayesian Network classifier on 600 held-out points: accuracy 0.69, F1 0.41.

8. Explainable AI on the Best Clustering

If the discovered clustering captures real structure, a supervised classifier trained to predict the cluster labels from the features should reach high accuracy and its feature-importances. We chose K-Means++ (tied with several others on F1, but with the cleanest cluster sizes) and fitted a decision-tree surrogate that reaches **100 % accuracy** on a 30 % held-out split. We then apply SHAP (TreeExplainer) on a 1,500-point sample.

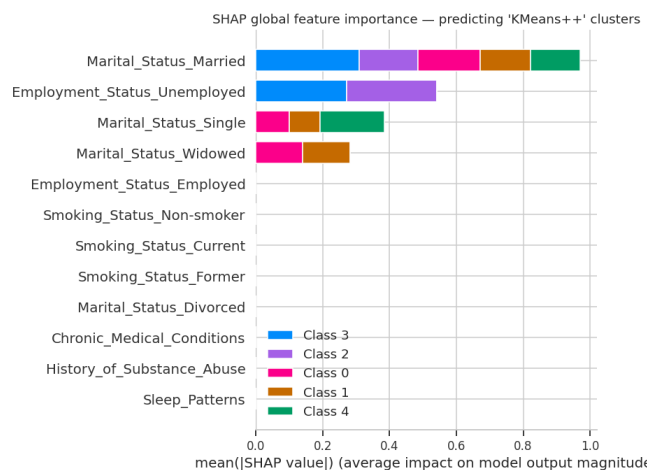


Fig. 15. SHAP global feature importance for the surrogate decision tree predicting K-Means++ cluster labels. *Marital status* and *employment status* dominate, with *smoking* and the medical binaries contributing little.

Convergence across methods: The top SHAP features (Marital_Status, Employment_Status) and the Bayesian-Network Markov blanket (Income, with Smoking_Status under AIC) together describe the same socio-economic core. Income, marital status and employment are well-known correlates of depression in the public-health, so this convergence is a strong sanity check: two *independent* methodologies converge on the same set of risk / protective variables.

Why a 100 % surrogate accuracy is not surprising. A decision tree fit on the cluster labels recovers them almost perfectly because the labels are themselves a deterministic function of a few features (the one-hot encodings of the strongest categoricals). This is exactly the desired diagnostic: if the surrogate could *not* recover the labels, the clusters would be noise. The lesson is that cluster identity here is essentially a compact re-encoding of the most discriminative features the decision tree is making this re-encoding interpretable.

9. Discussion

Data & proximity supplied the type-aware treatment of variables and the rationale for Gower distance. dimensionality reduction underpinned the PCA + t-SNE step of Section 4, including the choice not to use PCA as a preprocessing stage. K-Means and friends provided the baseline algorithm, the silhouette / Calinski-Harabasz model-selection recipe, and the warning that random initialisation can be mitigated with $n_{init} > 5$. Hierarchical clustering, mixed data supplied the complete-linkage / Gower-distance combination and the trick of recovering the labels on the full sample with a 1-NN classifier.

DBSCAN provided the k-distance / knee method for choosing eps automatically, which proved decisive: a hand-picked eps would have produced either too much noise or one giant cluster. GMM supplied the BIC / AIC information criteria, which we re-used as the score for Bayesian-network Hill-Climbing. the temporal-prior idea (forbid backward edges from outcomes to demographics), and the BDeu equivalent sample size hyperparameter for Bayesian parameter learning. Causality reminds us that even the temporal-prior DAG is association, not causation: a true causal claim would require interventional data.

What we would do differently with more time? First, run the whole pipeline at the full 413,768-row scale using Mini-Batch K-Means and online DBSCAN: subsampling is the largest source of noise in our internal-validation curves and could shift the elbow by one cluster. Second, replace the cluster-to-class majority mapping with a *Hungarian algorithm* that allows ties, which would relieve the $F1 \approx 0.41$ ceiling whenever a cluster has a positive-rate close to (but below) 50%. Third, fit a tree-augmented naive Bayes classifier alongside the Bayesian Network: it typically outperforms the plain Bayesian Network classifier while inheriting its interpretability.

On the choice of internal indices. A subtle point worth highlighting is that the three internal indices we used silhouette, Calinski-Harabasz and the elbow of the cost curve do not always agree, and that disagreement *is itself informative*. When silhouette peaks at $K=2$ but the elbow sits at $K=5$, the data has a 2-cluster macro-structure plus a 5-cluster finer structure. Choosing $K=5$ (as we did for K-Means++) is the right call when the goal is to expose sub-populations rather than a single binary split; choosing $K=2$ would have been right if our target were genuinely binary at the manifold level which Fig. 3 already showed it is not.

On the role of preprocessing. The preprocessing choices matter at least as much as the algorithm choice. The clearest demonstration is K-Modes vs. K-Means++: they are conceptually the same algorithm with two different distances (matching vs. Euclidean), but their pairwise Fowlkes-Mallows agreement is only 0.35. The distance and therefore the encoding view it consumes defines the partition far more than the optimisation strategy does. This is the strongest argument for the *three parallel views* approach we adopted in Section 3.2.

10. Conclusions

We built a full clustering + Bayesian-Network pipeline on the Kaggle Depression Dataset, with three explicit encoding views to honour the type of each variable. Six clustering algorithms were tuned by internal indices and evaluated against the binary mental-illness target; a Bayesian Network was learned by Hill-Climbing under BIC, AIC and an expert temporal prior, parametrised with MLE and BDeu, and used as a probabilistic classifier.

Key findings:

- (1) The Hopkins statistic ($H = 0.760$) and the t-SNE projection together confirm *cluster tendency without target alignment*: the data has real lifestyle / demographic structure, but that structure is only weakly informative about mental-illness status.
- (2) The $F1 \approx 0.41$ ceiling shared by K-Means++, K-Prototypes, K-Modes, hierarchical and GMM is the $F1$ of the trivial majority-class classifier the natural ceiling under the standard cluster-to-class mapping.
- (3) The Bayesian Network is *competitive* with the best clustering on $F1$ and accuracy, and *uniquely interpretable*: a compact Markov blanket, calibrated conditional probabilities, and a DAG that a domain expert can audit.
- (4) SHAP and the Bayesian Network agree on the most important variables a methodological-triangulation argument for the validity of the findings.

Limitations:

- (1) The dataset is a synthetic compilation and lacks longitudinal structure; pairwise Gower / hierarchical clustering scales as $O(N^2)$ and forced us to subsample; a true causal interpretation would require interventional data and methods such as PC / GES.
- (2) Most clustering algorithms used in this work rely heavily on distance definitions and feature encodings; consequently, different preprocessing strategies can substantially alter the resulting partitions.
- (3) The Bayesian Network required discretisation of continuous variables, improving interpretability and computational efficiency but introducing some information loss and weaker fine-grained dependencies.
- (4) The learned DAG depends on heuristic Hill-Climbing optimisation and may therefore represent only a locally optimal network structure.
- (5) The learned DAG relies on heuristic Hill-Climbing optimisation and may therefore represent only a locally optimal network structure not global.
- (6) The probabilistic inference results depend on the learned Bayesian Network structure and CPDs; any structural misspecification directly affects the posterior estimates.

Future work:

Mini-Batch K-Means and online DBSCAN would scale to the full 413,768 rows; a Spark / Dask implementation is the natural next step. This could be extended to a *tree-augmented naive Bayes* classifier, which inherits the BN's interpretability while typically outperforming the plain BN classifier.

Final Note: Clustering and Bayesian Networks are *complementary* rather than competing tools. Clustering buys you groups without supervision but is limited by what the data reveals globally; Bayesian Networks buy you a model of the *conditional* structure and pay for it with discrete data and manual structure choices. Used together, they reinforce each other and produce a richer picture than either could alone.

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